

DAT158 – Advanced algorithms

Exercise number 2

Notice, Problem 5 & 6 are part of the next Compulsory

Problem 1

Given $S = \{1,2,3,4,5\}$ and subsets

- $S_1 = \{1,2,5\}, w_1 = 2$
- $S_2 = \{1,3,4\}, w_2 = 4$
- $S_3 = \{2,5\}, w_3 = 5$
- $S_4 = \{2,4,5\}, w_4 = 6$
- $S_5 = \{3,4,5\}, w_5 = 8$
- $S_6 = \{4,5\}, w_6 = 3$

Use the Greedy algorithm to solve the set cover problem given. Slide 40 – 42, Chapter 1.

Problem 2

Given the following LP problem

$$\text{maximize } 4x_1 + 7x_2 + 6x_3$$

subject to

$$\begin{aligned}x_1 + 2x_2 + x_3 &\leq 7 \\ 2x_1 + x_2 + 3x_3 &\leq 14 \\ x_1 + 4x_2 + 2x_3 &\leq 11\end{aligned}$$

$$x_1, x_2, x_3 \geq 0$$

- What is the objective function (objektfunkson)?
- What are the decision variables (beslutningsvariabler / avgjerdsvariablar)?
- What are the constraints (betingelser/begrensninger)?
- Give two examples of infeasible solutions (ulovlege/umulige løysingar).
- Give two examples of feasible solutions (lovlege/mulige).

Problem 3

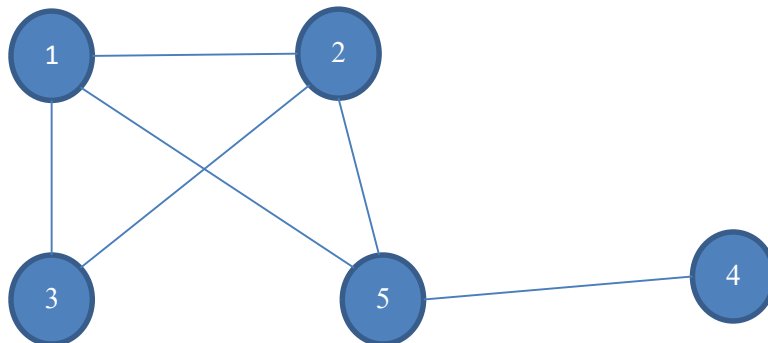
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- a) Formulate the set cover problem as an integer linear program (ILP) given the sets above.
- b) Formulate the LP relaxation of the problem in a)

Problem 4

A vertex cover of a graph is a subset of the vertices, $C \subseteq V$ such that every edge has at least one endpoint in C , that is $\forall (i,j) \in E$, either $i \in C$ or $j \in C$. We want to find a minimum vertex cover.



- a) Formulate the problem above as an integer linear program (ILP).
- b) Formulate the LP relaxation of the ILP above.

Problem 5 (part of Compulsory 2)

Given 7 jobs with corresponding processing times.

Job number	1	2	3	4	5	6	7
Processing time	4	7	2	9	12	5	3

The jobs will be processes on 3 identical machines without interruption. All jobs are available at time 0. (Model on slide 13, Chapter 2)

- Use the local search algorithm where we have all jobs on machine 1 in the beginning (in the same order as they are given). When you move a job, move it to the machine it can start earliest. If there is a tie, use the machine with the lowest number. What is the result, both time when the last job finish and the actual schedule?
- Use the Greedy algorithm on the same data. When have all jobs finished?
- Use the Longest processing time rule. When have all jobs finished?
- Give a lower bound on the best solution and give a short explanation?
- Is it possible to achieve the answer in d) (Explain)?

Problem 6 (part of Compulsory 2)

The adjacency matrix of a graph with 6 vertices is given below. The triangle inequality holds for the whole graph.

0	4	5	10	11	7
4	0	9	6	7	9
5	9	0	9	6	2
10	6	9	0	3	9
11	7	6	3	0	6
7	9	2	9	6	0

- Verify that the triangle inequality holds when going from vertex 1 to vertex 5.
- Run the nearest addition algorithm on the graph. If there are more than one candidate to add in a step, use the vertex with the smallest number. It is sufficient to have four nodes in the tour (skipping the last two iterations through the loop).
- Run the Double tree algorithm on the graph. If there are more than one minimum spanning tree, you should run it on every tree.
- Run Christofides' algorithm on the graph. If there are more than one minimum spanning tree, you should run it on every tree.